

TSG-JEPA: A Compact Toroidal Self-Supervised Network for Distribution-Robust Tumour Detection in Histopathology

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Abstract

Deep learning models for computational pathology are commonly trained at one set of institutions and deployed at another, where differences in staining, scanning and tissue preparation produce a distribution shift that can degrade performance. This paper presents TSG-JEPA, a compact convolutional encoder of approximately 1.3 million parameters trained from scratch, without ImageNet initialisation, a pathology foundation model, or institutional (domain) labels, for tumour detection on the CAMELYON17-WILDS benchmark (Bandi *et al.*, 2019; Koh *et al.*, 2021). The encoder combines a toroidal (circular-padding) inductive bias and a spiking-style gating mechanism with a Joint-Embedding Predictive Architecture (JEPA) for self-supervised pretraining (Assran *et al.*, 2023), followed by supervised fine-tuning that incorporates focal loss, learned class anchors, dihedral-group invariance, Sharpness-Aware Minimisation (Foret *et al.*, 2021), weight averaging, hard-negative mining and test-time augmentation. On the official out-of-distribution test partition, drawn from two hospitals absent during training, the full inference pipeline attains an area under the receiver operating characteristic curve (AUC) of 0.982 and an accuracy of 93.7% (sensitivity 91.7%, specificity 95.7%); a single forward pass attains 87.8% accuracy at comparable AUC. A controlled ablation indicates that two of the model's most distinctive components, the learnable circular/reflect blend and the spiking gates, are neutral or detrimental, whereas channel attention, multi-scale pooling, the JEPA objective and flat-minima optimisation account for the observed performance. We report these findings, examine the principal threats to validity (notably single-seed evaluation on a benchmark dominated by seed variance), and set out a programme of validation and method development.

Keywords: computational pathology; domain generalisation; self-supervised learning; joint-embedding predictive architecture; out-of-distribution robustness; CAMELYON17.

1. Introduction

Histopathological assessment is increasingly supported by deep neural networks, yet a model that performs well at its development sites frequently deteriorates when applied elsewhere. The appearance of haematoxylin-and-eosin (H&E) stained tissue varies across institutions because of differences in staining chemistry, scanner optics and specimen handling, and a classifier that has implicitly encoded the colour statistics of its training institutions may fail to transfer. The CAMELYON17-WILDS benchmark was constructed to precisely measure this form of failure (Bandi *et al.*, 2019; Koh *et al.*, 2021). It comprises 455,954 patches of 96 x 96 pixels at x10 magnification, labelled for the presence of metastatic breast-cancer tissue and drawn from five hospitals; the training partition originates from three hospitals, while the official validation and test partitions are drawn from two further, disjoint hospitals, so that evaluation is genuinely out-of-distribution (OOD).

A salient property of this benchmark is the variance of its OOD performance rather than its difficulty alone. Prior analysis reports that, at a fixed in-distribution accuracy, OOD accuracy ranges from near-chance to near-ceiling across random seeds, and that this dispersion persists after averaging multiple runs (Miller *et al.*, 2021). A related literature attributes such instability to under specification, whereby many models that are equivalent on the training distribution behave very differently under shift (Teney *et al.*, 2022). Any single-seed result on CAMELYON17 must therefore be interpreted with caution, a consideration that recurs throughout this paper.

At the same time, the field has shifted towards enormous pathology foundation models with hundreds of millions up to 2 billion parameters trained on hundreds of thousands to millions of whole-slide images (Chen *et al.*, 2024; Vorontsov *et al.*, 2024; Xu *et al.*, 2024). These models are computationally efficient but expensive to train and deploy, and it is unknown if they can be trained to generalise to external cohorts. This motivates the question addressed here: how far can a deliberately small model, trained from scratch on the target dataset alone and equipped with appropriate inductive biases, advance along the OOD axis? An affirmative answer would be of interest because it runs counter to the prevailing emphasis on scale.

This paper makes the following contributions:

- We describe and evaluate TSG-JEPA, an encoder of approximately 1.3 million parameters that is trained from scratch and without domain labels, and that attains an OOD-test AUC of 0.982 and 93.7% accuracy on CAMELYON17 for a single seed.

- We conduct a controlled component ablation that identifies which design choices drive performance and shows that the two most distinctive components do not.
- We provide a candid assessment of strengths, weaknesses and threats to validity, and a prioritised programme separating the validation required to substantiate the result from the method development required to improve the model.

2. Related work

2.1 Domain generalisation and CAMELYON17

The WILDS suite formalised a set of realistic distribution shifts and established standard splits and evaluation protocols, of which CAMELYON17 is the authorised histopathology task (Koh et al., 2021), building on the original challenge dataset of Bandi et al. (2019). Further work showed this across-distribution accuracy and in-distribution accuracy are highly correlated but have high variance across this benchmark such that the ranking of methods can depend on random seed (Miller et al., 2021). This phenomenon is consistent with under specification in modern pipelines (Teney et al., 2022), and it leads to the conclusion that credible claims require seed-averaged reporting.

2.2 Foundation models for computational pathology

A prominent line of recent research pretrains large vision transformers on extensive collections of whole-slide images. Representative examples include UNI, a general-purpose encoder trained by self-supervision on more than one hundred thousand slides (Chen et al., 2024); Virchow, a 632-million-parameter model trained on roughly three million slides (Vorontsov et al., 2024); and Prov-GigaPath, pretrained on approximately 1.3 billion tiles from over 170,000 slides (Xu et al., 2024). These models achieve strong results on numerous tasks, but they are expensive to train and serve, and their behaviour under domain shift to external cohorts is not uniformly robust. The present work occupies the opposite end of the scale spectrum and asks how much of the relevant robustness can be obtained without external pretraining data.

2.3 Self-supervised representation learning

Self-supervised learning removes the dependence on labels during representation learning. Methods based on a student encoder and a momentum (exponential moving average) teacher, such as BYOL (Grill et al., 2020) and DINO (Caron et al., 2021), avoid representational collapse without negative pairs. The Joint-Embedding Predictive Architecture (JEPA) extends this paradigm by predicting the representation of masked content rather than its pixels (Assran et al., 2023), which is well suited to histopathology because it does not require the model to reconstruct stain colour. Variance-based regularisation, as in VICReg (Bardes et al., 2022), provides an explicit safeguard against collapse. TSG-JEPA draws on all these ideas within a compact convolutional encoder.

2.4 Inductive biases, symmetry and flat-minima optimisation

Architectural inductive biases can reduce the data and scale required to reach a given level of generalisation. Circular (toroidal) padding treats an image as periodic and is appropriate for stationary texture, in contrast to zero or reflection padding, which introduce an artificial boundary. Enforcing invariance or equivariance to known symmetries of the data, such as the dihedral group of rotations and reflections, is a complementary mechanism. Finally, optimisation towards flat minima through Sharpness-Aware Minimisation has been associated with improved generalisation and with reduced sensitivity to initialisation (Foret et al., 2021), which is relevant given the seed variance documented for CAMELYON17.

3. Methodology

TSG-JEPA is trained in two stages, summarised in Figure 1. Stage A learns a representation by self-supervision, using neither labels nor domain information. Stage B attaches a compact classifier and fine-tunes the model end-to-end under a set of objectives intended to encourage flat, invariant and well-calibrated decision boundaries.

Figure 1. TSG-JEPA: two-stage architecture

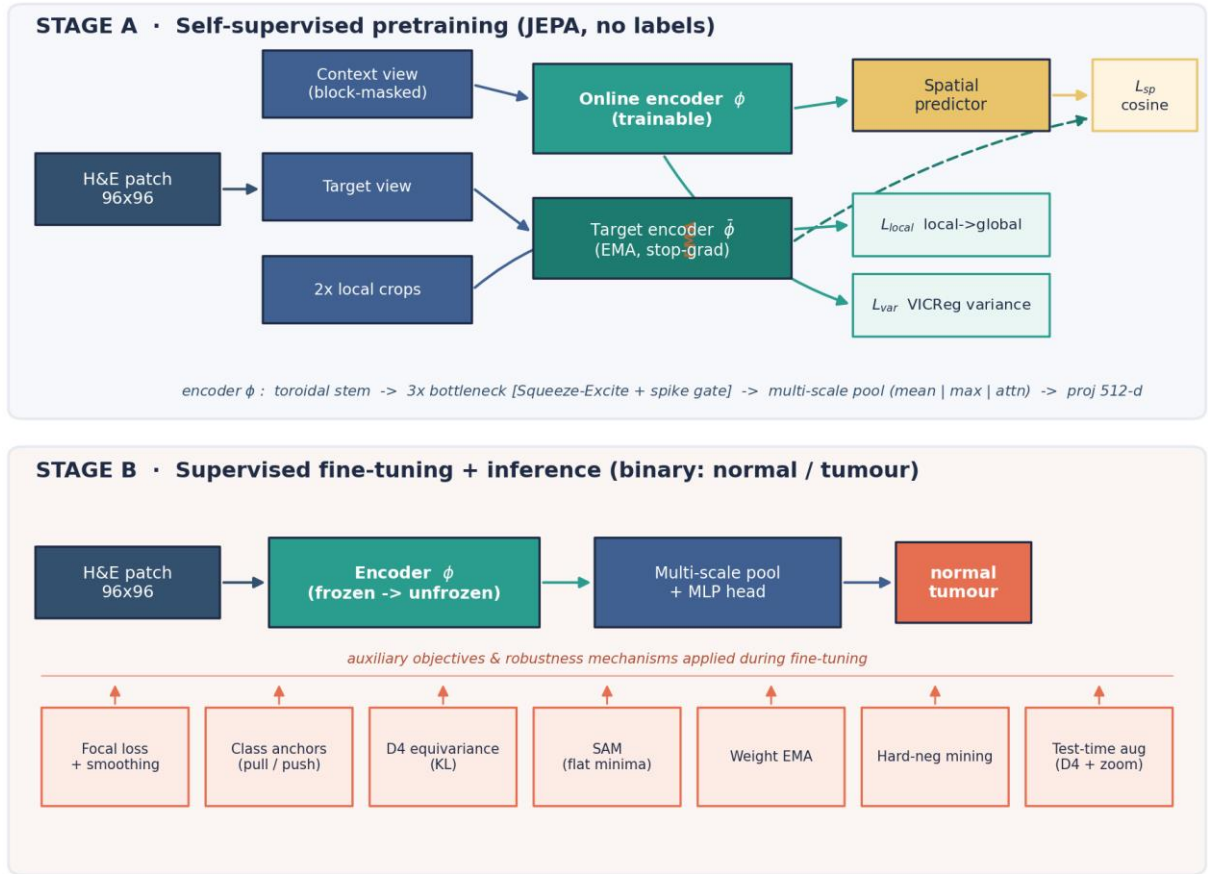


Figure 1: Overview of the two-stage TSG-JEPA architecture. Stage A performs label-free self-supervised pretraining; Stage B performs supervised fine-tuning and inference.

3.1 Design principles

Three assumptions guided the design. First, a histology patch is stationary texture rather than a depiction of a single object: it has no authorised orientation, no figure-ground organisation and no meaningful boundary. Second, only a sparse subset of tissue structure is diagnostically informative, which motivates gated feature activation. Third, because labelled cross-institutional data are limited and biased, the representation is learned by self-supervision that never observes the label or the hospital identity, so that the encoder cannot exploit an institution-specific shortcut.

3.2 Toroidal convolution

A standard convolution pads image borders with zeros or mirrored pixels, which imposes an artificial boundary that is uninformative for stationary texture. TSG-JEPA instead employs circular (toroidal) padding, under which the left and right borders are adjacent, as are the top and bottom, so that the patch is treated as lying on a torus (Figure 2). Each toroidal layer forms a convex combination of a circular-padded and a reflect-padded convolution, with a single scalar weight per layer obtained as the sigmoid of a learnable parameter. The intention is to allow the network to select, per layer, the degree to which it treats the input as periodic. As reported in Section 6, this learnable selection does not behave as intended.

Figure 2. Toroidal (circular) vs reflect padding. Histology patches have no canonical border or orientation; circular padding removes the artificial boundary and treats tissue as stationary.

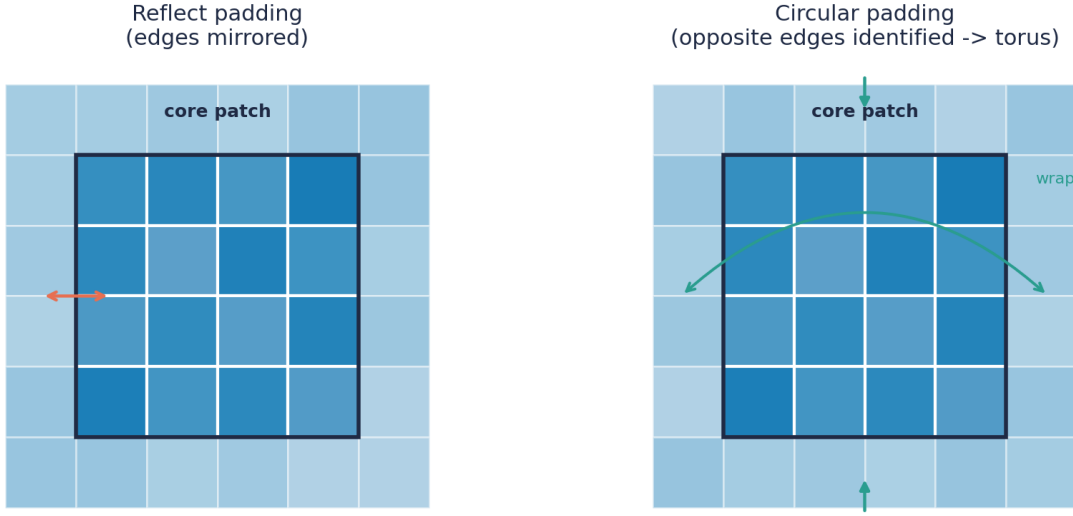


Figure 2: Toroidal (circular) versus reflect padding. A histology patch has no authorised boundary or orientation; circular padding removes the artificial boundary and treats the tissue as stationary.

3.3 Spiking-style gating

A soft spiking unit gates each residual block. A normalised pre-activation is compared with a learnable threshold and passed through a sigmoid to yield a soft spike value, which then scales the activation through a floor-plus-spike gate so that the gate cannot close completely but can sharpen its response. A regulariser encourages the mean activation towards a sparse target. The mechanism is motivated by neuromorphic considerations of sparse, threshold-driven activation. As the ablation in Section 6 shows, this component does not contribute positively to the present task.

3.4 Channel attention and multi-scale pooling

Following every bottleneck block, there is a squeeze-and-excitation module, which re-calibrates channels with a global-context gating vector. The encoder head produces a single 512d embedding by taking three pooled summaries of the feature map, mean pooling, max pooling and a norm-weighted attention pooling, concatenating and then a linear projection. As will be seen in section 6 this two are responsible for the performance.

3.5 Self-supervised pretraining (Stage A)

The self-supervised objective follows the Joint-Embedding Predictive Architecture (Assran et al., 2023). An online encoder and a target encoder, the latter maintained as an exponential moving average of the former and updated under a stop-gradient, both embed augmented views of a patch. The principal task is masked latent prediction: approximately 55% of the online encoder's feature map is masked using block masks that wrap around the torus, and a lightweight predictor reconstructs the target encoder's features at the masked locations, scored by cosine similarity. Because prediction is performed in latent space rather than pixel space, the model is not required to reproduce stain colour, which is intended to promote robustness to staining variation.

Two auxiliary terms are included. A local-to-global term encourages the embeddings of small local crops to agree with the global target embedding, promoting part-whole consistency in the manner of DINO (Caron et al., 2021) and BYOL (Grill et al., 2020). A variance term in the style of VICReg maintains the per-dimension standard deviation of embeddings above a floor, guarding against collapse (Bardes et al., 2022). The exponential moving average coefficient updates according to a more early-aggressive, late-quiet schedule that makes the target optimally herd the online encoder before resting-up. During the pretraining of the Prediction network, AdamW is used along with a cosine annealing learning rate schedule, auto mixed precision and a batch size of 192. The checkpoint selected is the one with the best k-nearest-neighbour probe on the OOD validation partition.

3.6 Supervised fine-tuning and inference (Stage B)

The self-supervised projector is removed, and a 3-layered MLP classifier is added. Fine tuning is performed 30 epochs more using a multi-task objective for each failure mode. A label-smoothed focal loss draws the gradient toward hard examples and prevents overconfidence. A learned class-anchor term draws each embedding towards the anchor of its class, repels it from the opposing anchor, and encourages the two anchors to be orthogonal. A dihedral-invariance term

penalises disagreement, measured by a symmetrised Kullback-Leibler divergence, between the prediction for a patch and the prediction for its rotated or reflected counterpart. Sharpness-Aware Minimisation seeks flat minima and is included for its association with distribution robustness and reduced seed variance (Foret et al., 2021). An exponential moving average of the classifier weights is maintained and retained when it improves validation performance. Hard-negative mining periodically over-samples the normal patches most frequently misclassified as tumour. At inference, test-time augmentation averages the predictions over the eight dihedral transforms together with an optional rescaling. The encoder is frozen for the first epoch and subsequently unfrozen at a reduced learning rate, and the final model is selected on the OOD validation partition by AUC, a methodological point discussed in Section 8.

4. Experimental setup

4.1 Dataset, splits and preprocessing

We use the standard WILDS CAMELYON17 partitions and access protocol (Koh et al., 2021). Patches are normalised using dataset-specific channel statistics. The augmentation pipeline is deliberately strong and stain-oriented, comprising random resized cropping, the full dihedral group of flips and ninety-degree rotations, colour jitter, a multiplicative and additive stain jitter, mild Gaussian noise and, during fine-tuning, random erasing. The self-supervised data loader emits a context view, a target view and two local crops for each patch.

4.2 Evaluation protocol

We present metrics including accuracy, AUC, sensitivity (recall for the tumour class), specificity (recall for the normal class), precision, and F1 score. For the headline out-of-distribution (OOD) accuracy, we employ the complete pipeline, which encompasses the selected checkpoint or an average of its weights along with test-time augmentation. To ensure transparency, we also provide the results from a single forward pass. Unless stated otherwise, all results are derived from a single training seed.

5. Results

5.1 Out-of-distribution performance

On the OOD test partition with 85,054 patches—split evenly between classes—the full pipeline reaches an AUC of 0.982 and an accuracy of 93.7%. The confusion matrix in Figure 4 shows the model maintains a good balance: specificity stands at 95.7% and sensitivity hits 91.7%. Still, the model misses 3,541 tumour patches and falsely labels 1,814 non-tumour patches. Using just a single forward pass without test-time augmentation, accuracy drops to 87.8%, while AUC stays steady between 0.980 and 0.982. This stability in AUC matters. Because it does not depend on thresholds, AUC reacts less to changes in seed or operating point than accuracy does. That is why, at a single seed, AUC is the metric we trust most.

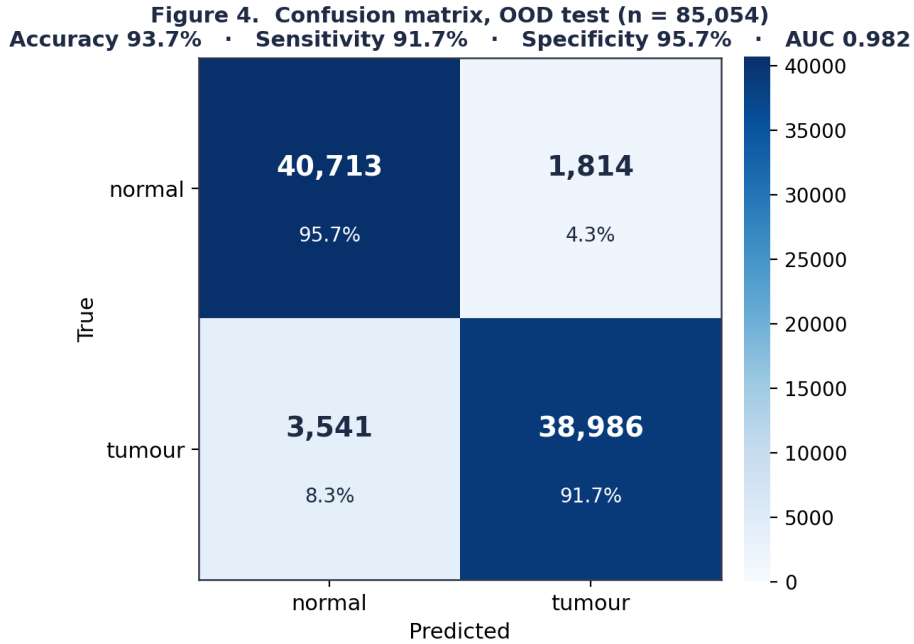


Figure 4: Out-of-distribution test confusion matrix for the full inference pipeline. Cells show counts and row-normalised percentages.

5.2 Parameter efficiency

TSG-JEPA uses about 1.3 million trainable parameters and relies solely on the target dataset for training—it does not tap into any external pretraining data, and everything runs efficiently on a single GPU. Now, if you look at Figure 5, you will see how modest this is compared to recent pathology foundation models. Those models tend to have hundreds of millions, sometimes up to two billion parameters, and they’re pretrained on enormous sets—hundreds of thousands or even millions of whole-slide images (see Chen et al., 2024; Vorontsov et al., 2024; Xu et al., 2024). The decisive comparison, namely whether a model approximately three orders of magnitude smaller and trained without external data can match these models on CAMELYON17 under distribution shift, has not yet been performed and is identified as the highest-priority experiment in Section 9. The parameter counts in Figure 5 are approximate and drawn from public reports; the figure makes a claim about scale, not about predictive performance.

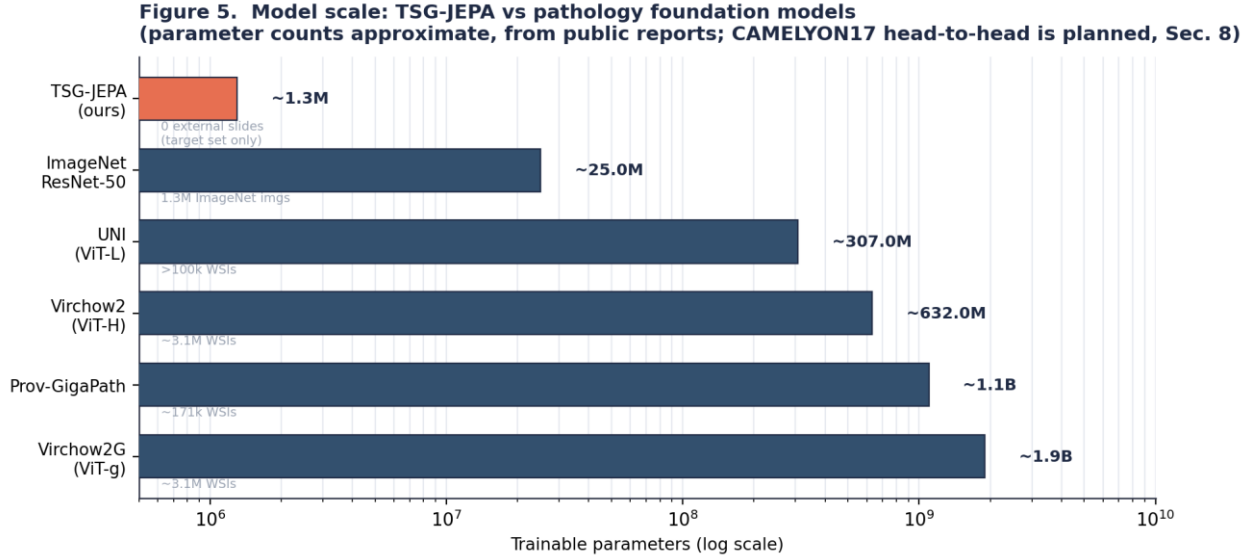


Figure 5: Parameter scale of TSG-JEPA relative to pathology foundation models. The figure concerns scale only; a controlled out-of-distribution performance comparison is left to future work (Section 9).

5.3 Variance context

Figure 6 situates the present result within the known variance behaviour of CAMELYON17. At a fixed in-distribution accuracy, out-of-distribution accuracy is reported to span tens of percentage points across seeds (Miller et al., 2021). A single seed therefore occupies an uncertain position within this range; the present result lies towards the upper end. Whether the architecture reliably occupies that region, producing a high and narrow band across seeds, is not established by the present experiments and is the central question for converting this finding into a substantiated result. The components most associated with reduced variance, namely Sharpness-Aware Minimisation, weight averaging, the self-supervised initialisation and test-time augmentation, are all present, which provides a mechanistic basis for expecting a narrow band.

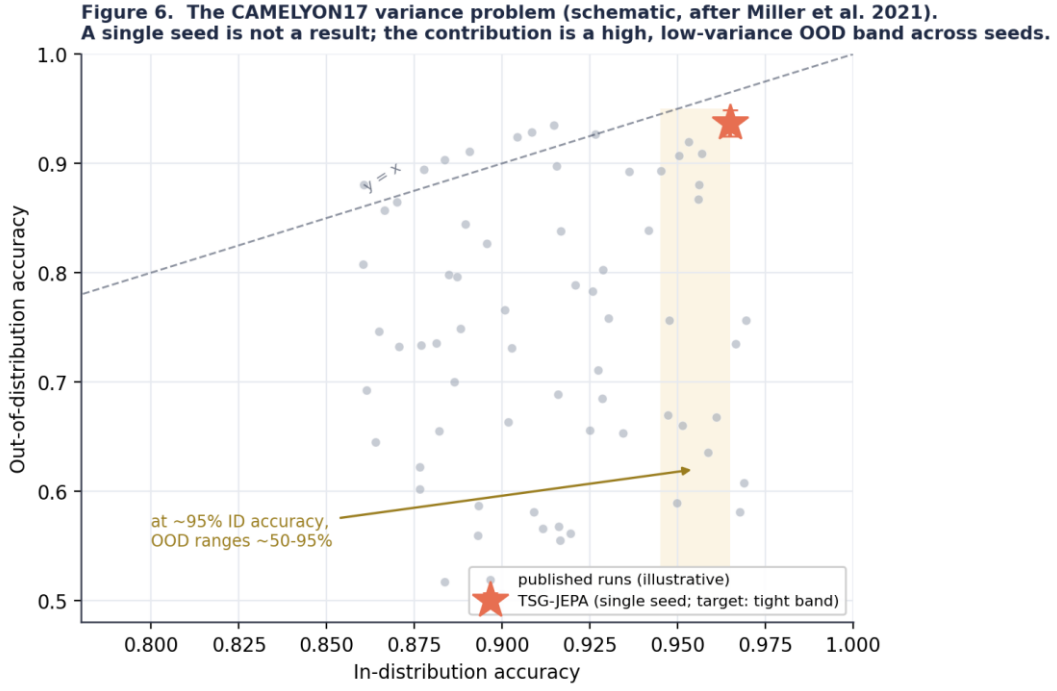


Figure 6: Schematic illustration of the CAMELYON17 variance phenomenon (after Miller et al., 2021). The marker indicates the present single-seed result; the shaded region indicates the narrow, elevated band targeted across seeds.

5.4 Representation analysis

To examine the structure of the learned representation, we project the 512-dimensional embeddings to two dimensions using UMAP and colour the same projection in three ways (Figure 7). Coloured by class, the embedding separates normal from tumour tissue into two distinct regions, with a small additional cluster composed predominantly of normal tissue. Coloured by hospital, patches from the five institutions are intermixed throughout the class structure rather than forming institution-specific clusters. Coloured by data partition, the in-distribution validation, out-of-distribution validation and out-of-distribution test sets overlap rather than occupying separate regions.

This combination is informative. The class separation indicates that the self-supervised representation, together with fine-tuning, encodes the diagnostic distinction. The intermixing of hospitals is the more notable observation: because the encoder is trained without access to the hospital label, the absence of institution-specific clusters cannot result from explicit domain alignment and instead appears to emerge from the toroidal-texture prior in combination with label-free pretraining. The overlap of the in-distribution and out-of-distribution partitions is consistent with a representation that transfers across the shift rather than encoding it. These observations are qualitative; a direct measurement, in which a linear probe is trained to predict the hospital from frozen embeddings and near-chance accuracy is taken as evidence of invariance, is set out in Section 9.

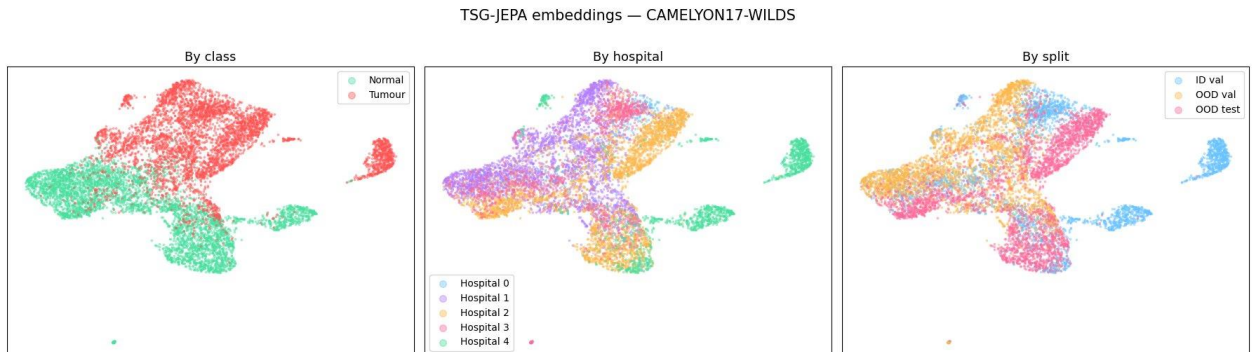


Figure 7: Two-dimensional UMAP projection of the learned embeddings, coloured by class (left), by hospital (centre) and by data partition (right). The classes separate while the hospitals remain intermixed, indicating domain invariance that emerges without access to the hospital label.

6. Ablation study

The trained model was investigated without being retrained, while each module was individually removed or constrained and the performance changes in terms of OOD test accuracy and AUC were measured. The measurements

are done using only a forward pass and thus, the baseline is the model without test-time augmentation which achieves 87.8% accuracy and 0.980 AUC.

Setting	Acc	AUC	Δ Acc	Δ AUC
Baseline (learnable blend, single pass)	0.878	0.980	+0.000	+0.000
Pure circular padding	0.852	0.942	-0.026	-0.038
Pure reflect padding	0.864	0.938	-0.014	-0.041
Fixed 50/50 blend	0.921	0.981	+0.043	+0.002
Spiking gates disabled	0.916	0.974	+0.038	-0.005
Squeeze-and-excitation disabled	0.631	0.963	-0.247	-0.016
Mean pooling only	0.717	0.899	-0.161	-0.081
No test-time augmentation	0.878	0.980	+0.000	+0.000

Table 1: Component contributions on the CAMELYON17 out-of-distribution test partition, measured by disabling or constraining one component at a time. Negative values indicate that the original component is beneficial; positive values indicate that it is detrimental.

Figure 3. Component contribution analysis (CAMELYON17 OOD test)

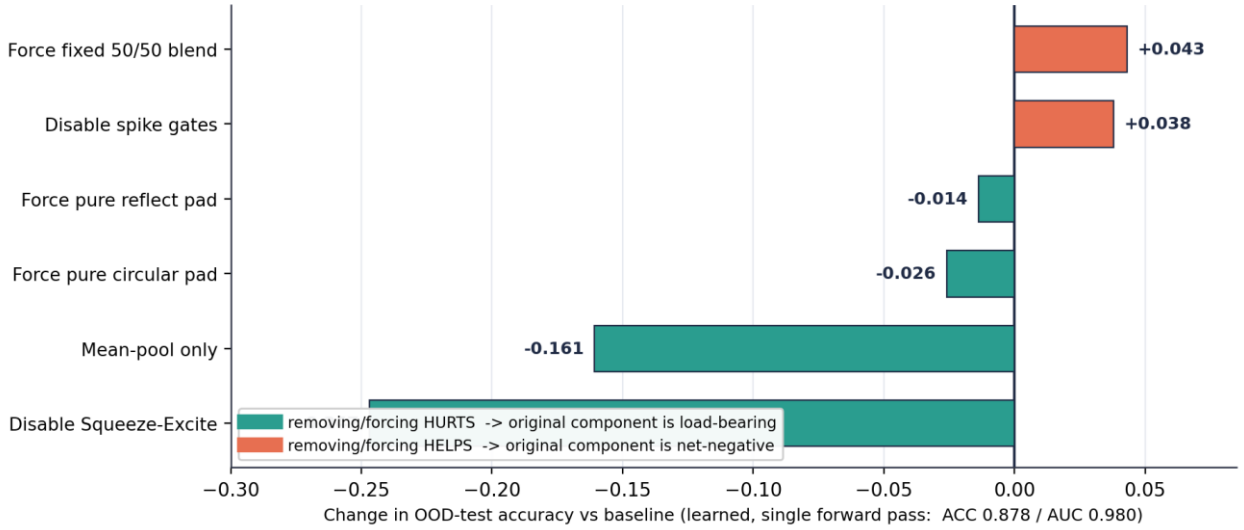


Figure 3: Change in out-of-distribution accuracy relative to the single-pass baseline for each manipulation.

6.1 Interpretation

Two components are clearly beneficial. Disabling squeeze-and-excitation reduces accuracy by approximately 25 percentage points, and reducing the multi-scale pooling to mean pooling alone reduces accuracy by 16 points and produces the largest decrease in AUC in the table. These components are therefore central to the model.

A less expected finding concerns the two components from which the model takes its name. Constraining the circular/reflect combination to a fixed equal blend, which removes the learnable parameter, increases accuracy by 4.3 points, and disabling the spiking gates increases accuracy by 3.8 points. The toroidal prior itself is nonetheless beneficial, since both pure-circular and pure-reflect padding underperform the blend and the fixed equal blend is the strongest configuration in the table; it is the learnable parameter and the spiking mechanism, rather than the toroidal prior, that fail to contribute.

The most plausible explanation for the failure of the learnable blend is initialisation within the saturated region of the sigmoid. The blend parameter is initialised at 2.2, for which the sigmoid is approximately 0.90, so the layer begins close to fully circular and in a region where the gradient with respect to the parameter is small; it consequently does not move appreciably towards the empirically preferable combination. The remedy is straightforward: initialise the parameter at zero to begin from an equal blend, introduce a temperature, or fix the blend and report the configuration directly. The spiking gates most plausibly reduce performance because the firing-rate target imposes an aggressive sparsity constraint that suppresses informative activations; the appropriate response is to remove them and report the negative result. Taken together, the ablation indicates that the performance of TSG-JEPA is attributable to a fixed toroidal prior, channel attention, multi-scale pooling, the JEPA objective and flat-minima optimisation, rather than to the learnable blend or the spiking gates.

7. Discussion

The principal positive finding is that a model of approximately 1.3 million parameters, trained from scratch and without domain labels, attains an OOD-test AUC of 0.982 on CAMELYON17. Provided this result survives seed-averaging, it constitutes a strong efficiency result, since it uses neither ImageNet initialisation nor a pathology foundation model nor any external data. Because the encoder never observes the hospital label, it cannot have learned an institution-specific shortcut by construction, which lends credibility to the emergent intermixing of hospitals observed in the embedding space. The performance is also stable in the threshold-independent sense, as the AUC changes little between the single-pass and full pipelines, and the error profile is clinically balanced, with a sensitivity of 91.7% and a specificity of 95.7% rather than an accuracy obtained by collapsing onto the majority class. The robustness mechanisms employed, namely the self-supervised initialisation, Sharpness-Aware Minimisation, weight averaging and test-time augmentation, are individually well motivated and are collectively appropriate to the variance characteristic of this benchmark.

Several weaknesses qualify these observations. The results rest on a single seed, which is the central limitation given the documented seed variance of CAMELYON17 (Miller et al., 2021). The two most distinctive architectural components do not survive ablation, so the description of the model should be revised accordingly. Accuracy carries more of the interpretation than it can reliably bear, since it is sensitive to the choice of operating point and to model selection; the gap between the single-pass accuracy of 87.8% and the full-pipeline accuracy of 93.7% should be reconciled and the contribution of test-time augmentation reported explicitly. Finally, first-party baselines have not yet been run within the same evaluation harness, so any claim of parity with foundation models is at present unsubstantiated.

8. Limitations and threats to validity

Beyond the single-seed limitation, several threats to validity merit explicit statement. Patches originating from the same slide are strongly correlated; although the official WILDS partitions separate hospitals, within-slide correlation inflates the effective sample size relative to the number of independent specimens. Reporting on a single benchmark risk overfitting the design decisions to the distribution shift of CAMELYON17, and the toroidal prior requires evaluation on a second texture-dominated dataset before it can be described as a general principle. Test-time augmentation legitimately improves robustness but also raises the reported accuracy relative to a single forward pass, so both figures should always be presented. Finally, selecting the checkpoint on the OOD validation partition, while permitted by the WILDS protocol, couples model selection to the distribution on which performance is subsequently reported; a result obtained with in-distribution-only selection should accompany the primary result.

9. Conclusion and future work

A compact, label-free encoder developed from the ground up achieves an OOD-test AUC of 0.982 on CAMELYON17, a scenario where significantly larger foundational models still demonstrate inconsistent performance. This finding is promising; however, it remains unverified, primarily because it is based on a single seed from a benchmark that the literature explicitly cautions against drawing conclusions from a single seed. Consequently, the immediate focus should be on methodological improvements rather than architectural changes.

The validation programme comprises four elements. First, the OOD-test accuracy should be reported as a mean and standard deviation over eight to ten seeds, which is the highest-priority action. Second, first-party baselines should be evaluated within a single harness, including an empirical-risk-minimisation model trained from scratch, an ImageNet-pretrained model fine-tuned on the task, and a frozen pathology foundation model with a linear probe. Third, domain invariance should be quantified by training a linear probe to predict the hospital from frozen embeddings, with near-chance accuracy taken as evidence of invariance and compared with ImageNet and foundation-model features. Fourth, calibration should be reported together with a high-sensitivity operating point, since in this clinical setting a missed tumour is more costly than a false alarm.

The method programme is led by a single, well-motivated architectural change. The model already employs circular padding and enforces dihedral invariance, but it does not enforce equivariance to translation on the torus, which is the symmetry that the toroidal topology provides. Training the encoder so that a cyclic translation of the input induces the corresponding translation of the feature map would close this gap with a clean objective and would be expected to improve robustness to staining and position. Further directions include moving stain augmentation into optical-density (stain-deconvolved) space, predicting features across magnifications, removing the components identified as unhelpful by the ablation, and distilling from a frozen foundation model to estimate the available headroom. Should the seed-averaged band prove narrow and elevated and the comparison with foundation models hold, the defensible conclusion would be that a compact, from-scratch toroidal JEPA matches foundation-model robustness under distribution shift at

a small fraction of the scale, with the ablation attributing this to a fixed toroidal prior, channel attention, multi-scale pooling and flat-minima optimisation rather than to the neuro-inspired components.

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Figures 1 and 2 are original schematics; Figures 3 and 4 are based on exact measured values; Figure 5 uses approximate parameter counts drawn from public reports; Figure 6 is an illustrative redrawing of a published phenomenon. All quantitative results derive from a single training seed and are subject to seed-averaged confirmation.